

Enhancing Accommodative Function in XR via Dynamically Modulated Peripheral Chromatic Blur

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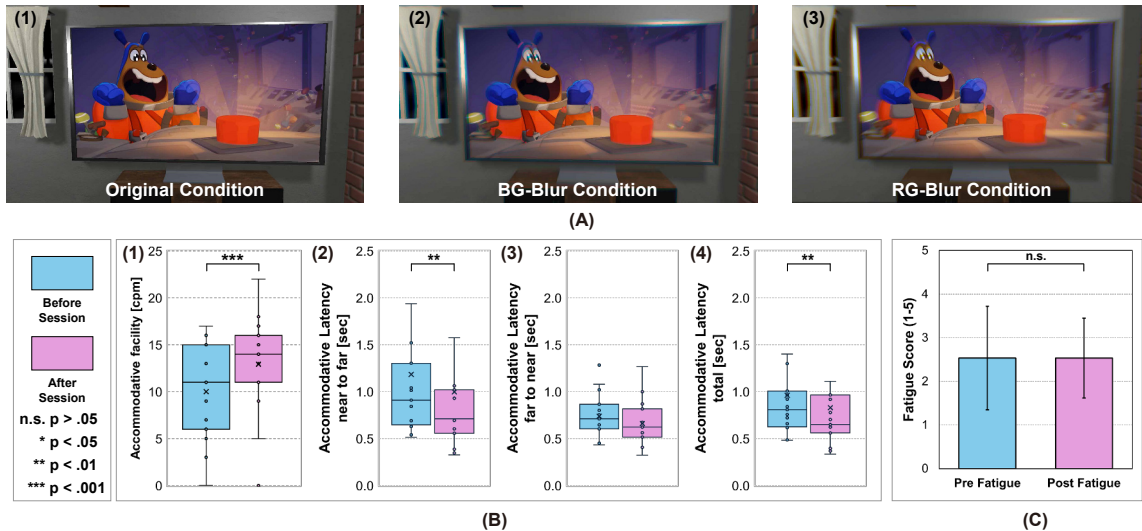


Figure 1: Overview of the proposed peripheral chromatic-defocus stimulus and outcome measures. (A) Example Stimulus shown in the virtual viewing environment under three rendering conditions: (1) Original Condition (no peripheral blur), (2) BG-Blur Condition (simulating myopic defocus cue), (3) RG-Blur Condition (simulating hyperopic defocus cue). (B) Pre- and post-session comparisons of accommodative function: (1) Accommodative facility [cycle per min, cpm], (2) Near-to-far (NF) accommodative latency [sec], (3) Far-to-near (FN) accommodative latency [sec], (4) Total accommodative latency [sec; NF+FN]. (C) Questionnaire results for subjective visual fatigue scores (1-5 Likert; higher = worse symptoms): Pre fatigue (assessed before the session), Post fatigue (assessed after the session).

ABSTRACT

Extended reality (XR) displays require users to view content at near distances for extended periods, leading to sustained ciliary contraction that may induce eye fatigue, pseudomyopia, and presbyopia. We propose a method that dynamically alternates peripheral chromatic defocus blur based on the principle of longitudinal chromatic aberration, aiming to train and enhance accommodative function during XR use without users perceiving the blur. We conducted refraction and accommodative tests, along with subjective assessments, to evaluate this approach. The method improved accommodative facility and reduced near-to-far latency, suggesting that dynamic chromatic blur can enhance accommodative function without being perceptually noticeable to users.

Index Terms: Extended Reality, Accommodative Function, Longitudinal Chromatic Aberration, Peripheral Vision, Visual Fatigue, Pseudomyopia, Presbyopia.

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1 INTRODUCTION

As the use of extended reality (XR) systems increases across diverse fields, users are increasingly exposed to prolonged viewing with a fixed focal distance. In current fixed-focus head-mounted displays, stereoscopic rendering presents virtual objects at varying perceived depths while the optical focal plane remains constant, creating a vergence-accommodation conflict (VAC). Although no accommodative change is optically required, the neural cross-link between vergence and accommodation induces the ciliary muscle to attempt adjustments that receive no confirmatory blur feedback. This persistent mismatch generates chronic control errors, leading to ciliary strain and reduced accommodative flexibility over time [1]. However, most existing interventions remain largely passive and do not directly engage accommodative function.

Longitudinal chromatic aberration (LCA) provides a physiological cue for defocus direction, as different wavelengths focus at distinct dioptric planes. Prior work has shown that chromatic aberration rendering can drive accommodation effectively [2], and recent findings highlight retinal mechanisms linking LCA-consistent defocus cues to eye growth regulation [3]. Building on these findings, we investigate whether peripheral chromatic blur can be used to train accommodative function in XR. By dynamically changing chromatic blur to the peripheral visual field, we aim to engage accommodative dynamics without interfering with user experience.

2 METHODOLOGY

2.1 Participants

A total of 15 participants (11 male; mean age: 25.1 ± 2.0) were recruited for this study. All participants underwent autorefractometry examination with naked eyes to assess baseline refractive status. For accommodative function tests and the experimental session, all participants used their habitual visual correction (glasses or contact lenses) to reflect real-world usage.

2.2 Apparatus and XR Environment

The XR environment was developed in Unity (6000.0.23f1) using a custom shader for gaze-contingent rendering. Participants wore a Varjo XR-4 HMD with integrated eye tracking (200 Hz), which provided real-time gaze point data. Refractive characteristics were measured using an autorefractometer (WAM-5500, Shigiya Machinery Works Ltd.). Accommodative facility was assessed using ± 2.0 D flipper lenses at 40 cm, and accommodative latency was measured using targets positioned at 40 cm and 3 m. All measurements were performed with the participant's head stabilized.

A realistic indoor VR environment was implemented for the task, with a virtual TV positioned 1.5 m away displaying video content. Participants remained seated with stable posture but were free to look around the environment without gaze restriction.

2.3 Stimulus Design and Implementation

To apply dynamic peripheral LCA blur at a level below participants' conscious awareness, we first derived JOD (Just-Objectionable Difference) thresholds from pairwise comparisons in a pilot study. These thresholds were used to determine the peripheral field-of-view (FOV) boundary and the LCA blur strength (Gaussian σ values) that remained below the perceptual detection threshold.

Foveal and peripheral regions were defined in angular terms using real-time eye tracking. For each frame, the system calculated the gaze point and corresponding viewing angle as a function of viewing distance, and these values were passed to the shader. The FOV criterion was defined at a viewing distance of 1.5 m, with the foveal region set to $\leq 45^\circ$ and the peripheral region to $> 45^\circ$, and rescaled according to a distance-dependent conversion factor.

Based on the per-frame FOV information, the shader separated foveal and peripheral regions and applied chromatic blur only to the periphery. To minimize awareness of the transition, a smooth alpha-blending ramp (0–1) was applied across the boundary.

LCA blur was implemented using Gaussian filtering in RGB space. Each frame was decomposed into R, G, and B channels, a channel-specific blur was applied, and the channels were then recombined to form the final image. The effective blur strength was determined by the selected σ values. Finally, to create a dynamic accommodative training stimulus, blue–green and red–green blur patterns were alternated using a sinusoidal transition with a 5 s period (0.2 Hz), a temporal frequency commonly used in dynamic accommodation paradigms (see Fig. 1A).

2.4 Experimental procedure and Outcome measures

The experimental session consisted of four consecutive phases: (1) **Pre-test** — Baseline measurements of visual fatigue, refraction, and accommodative function; (2) **Exposure** — a 45-minute video viewing session with dynamic peripheral LCA blur applied; (3) **Post-test** — identical measurements to the pre-test; and (4) **Questionnaire** — assessment of visual fatigue.

Refraction was measured at both near (40 cm) and far (3 m) distances. Accommodative response was calculated as $SE_{\text{far}} - SE_{\text{near}}$, and accommodative lag as $2.17 \text{ D} - \text{Response}$, where 2.17 D represents the dioptric demand difference between 40 cm (2.50 D) and 3 m (0.33 D). Accommodative facility was expressed as cycles per minute. Total latency was computed as the sum of near-to-far (NF) and far-to-near (FN) latencies.

Visual fatigue was assessed using a 5-point Likert-scale questionnaire (1 = strongly disagree, 5 = strongly agree). A single baseline item (Pre visual fatigue) was administered before the pre-test measurements. After the test, participants completed a post-session fatigue item (Post visual Fatigue).

All refraction and accommodation assessments followed standard optometric protocols and were reviewed by a licensed optometrist.

3 RESULTS

Data met normality assumptions (Shapiro–Wilk), and a paired t -test was conducted to compare accommodative facility between the pre- and post-task conditions. The results showed that accommodative facility significantly improved from pre- to post-task ($t(14) = 5.28$, $p < .001$, Cohen's $d = 1.36$; see Fig. 1B).

Next, paired t -tests were applied to compare accommodative latency between pre- and post-task conditions. Results showed that NF latency significantly decreased from pre- to post-task ($t(14) = -3.49$, $p = .004$, $d = -0.90$). In addition, FN latency showed a trend toward improvement from pre- to post-task ($t(14) = -1.88$, $p = .081$, $d = -0.49$), and total latency also decreased significantly by 13.5% from pre- to post-task ($t(14) = -3.08$, $p = .008$, $d = -0.80$).

Next, to investigate latency changes influenced by transition direction, a repeated-measures ANOVA was conducted to examine the effects of Time and Direction. Results showed a significant main effect of Time ($p = .004$) and a significant Time $\times$ Direction interaction ($p = .020$), indicating that latency reduction was greater for NF than FN transitions. Accommodative lag showed no significant change ($t(14) = -0.96$, $p = .352$, $d = -0.25$).

Finally, a paired t -test was conducted to investigate whether dynamic peripheral chromatic blur induced visual fatigue. Results showed no significant difference between pre- and post-task fatigue scores ($t(14) = 0.00$, $p = 1.00$, $d = 0.00$; see Fig. 1C).

4 DISCUSSION

Dynamic peripheral chromatic blur grounded in LCA principles significantly enhanced accommodative function during a single 45-minute XR session. The large effect sizes observed for accommodative facility ($d = 1.36$) and NF latency ($d = -0.90$) indicate clinically meaningful improvements in ciliary muscle flexibility, relevant to both pseudomyopia prevention and presbyopia management [4]. The significant Time $\times$ Direction interaction suggests that NF transitions—relaxation from the accommodated state—benefited more than FN transitions. This asymmetry may be attributed to the young adult sample (mean age 25.1 years), whose accommodative capacity remains sufficient for near focusing but may benefit more from relaxation training. The low subjective discomfort ratings and unchanged fatigue levels confirm that the peripheral chromatic blur was well-tolerated during extended viewing. Overall, these results suggest that XR displays can actively support visual health through LCA-based peripheral blur training, achieving significant improvements in accommodative function (+29.3% facility, -13.5% total latency) with minimal discomfort.

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